PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM TWO - MOCK DISTRICT/REGION - EXAMINATION BIOLOGY

MARKING SCHEME.

SECTION A (15 MARKS)

1.

(I)	(II)	(III)	(IV)	(V)	(VI)	(VII)	(VIII)	(IX)	X
C	C	A	A	B	C	D	D	D	B

1 mark @ = 10 marks

2.

List A	(I)	(II)	(III)	(IV)	V
List B	G	A	D	E	C

1 mark @ = 05 marks.

(a) A Student is required to explain five - Characteristics of living things, which make/support worms to be living things.

(i) They can grow
(ii) They can move and locomote. eg Earthworm
(iii) They can undergo excretion.
(iv) They can sense.
(v) They can reproduce
(vi) They can respire
(vii) They can feed.

Any five points 1@mark = 5marks.

(b) A student is required to give out five - importance of Studying Biology

(i) It helps to understand and care for the environment.
(ii) It helps to identify and group living things which makes it easier to learn about them.
(iii) It helps to appriciate nature and many fascinating things about different organisms and how their functions are learnt in Biology.
(iv) It helps to understand our bodies.
(v) It helps to acquire research skills
(vi) It helps to improve food production.

Any five related points = 1mark@ = 5marks.

(a) Distinctive features of Angiosperms.

(i) They are flowering plants.
(ii) Their seeds are enclosed in an ovary.
(iii) They produce seed bearing fruits.
(iv) They have well developed root system, stem and leaves.
(v) Angiosperms have well developed conducting tissues i.e Xylem and phloem.

Any four features, 1 mark @ = 4 marks

(b) Scientific names

(i) Lion - - - - Panthera leo.
(ii) Human being - - - Homo sapiens
(iii) Maize - - - - Zea mays
(iv) Housefly - - - - Musca domestica
(v) Domestic cat - - - Felis catus
(vi) Mimosa plant - - - Mimosa pudica.

@ 1 mark = 5 marks.

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@ 1 mark = 5 marks.

(a) Difference between an owl (animal) cell and maize (plant) cell.

An Owl	Maize cell
(i) Has no chloroplast	Has chloroplast.
(ii) Lacks cell wall	Has cell wall
(iii) It is irregular in shape	It is regular in shape.
(iv) Has small temporary vacuole.	Has large permanent vacuole.
(v) Stores food in forms of glycogen	stores food in forms of starch.
(vi) Has centrioles	Has no centrioles.
(vii) Nucleus is centrally positioned.	Nucleus is positioned at the peripheral.

Any five points 01 mark @ = 05 Marks.

(b) Components of cell theory

(i) All living things are composed of one or more cells.
(ii) Cells are the basic unit of structure and functions of living things.
(iii) All cells are produced from other cells
(iv) Cells contain inheritable information which controls their activities
(v) All cells are basically similar in chemical composition.
(vi) All life processes take place in the cell.

Any five points 01 mark @ = 05 marks

(a) (i) Raw Materials
(i) Water
(ii) Carbon dioxide. 02 marks

(ii) Condition necessary
(i) Sunlight
(ii) chloroplyll 02 marks

(b) Chemical equation of photosynthesis

$6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6H_2O + 6O_2$

02 marks for a well balanced equation

(c) Importance of photosynthesis
(i) It helps in production of oxygen.
(ii) It helps in Reduction of Atmospheric carbondioxide.
(iii) production of food.
(iv) It helps in Conversion of solar energy into chemical energy.

Any 4 point 1mark@ = 04 marks

TOPICS TO BE COVERED

FORM I

Nutrition
Classification
Introduction
Cell
Laboratory rules
First Aid.

FORM II

(a) A Student is required to draw a well labeled of human digestive system

Heading = 01
Diagram = 02
Any four labelling ors@ = 02
Total = 05 marks

(b) (i) Ileum = 01 mark

(ii) Adaptations of Ileum to its functions

(a) The Ileum has secretory gland that secret digestive enzymes.

(b) It contain mucus that ensures - Constant lubrication of internal walls for effective digestion of food.

(c) It is long about six meters to ensure maximum absorption of the end product of digestion.

(d) The inner wall of Ileum has - many villi to increase surface area for absorption.

(e) It is supplied with many blood - capillaries to increase surface - area for absorption of end product of digestion.

Any four points, 01@ marks = 04

(i) Virus

Is an extremely small fragment of nucleic acid (RNA or ~~DNA~~)

While

02 marks

Bacteria

Are microscopic organism that live nearly every environment on earth.

Or any meaningful Difference

(ii) Artificial and Natural System of classification

Artificial System	Natural System
(i) Consider few features	Consider many features
(ii) Require simple skills	Require advanced scientific skills.
(iii) It's less expensive	It's expensive.

Any one difference 02 marks

(iii) Biology laboratory and school library.

Biology laboratory	School library
(i) Has Biological models	Has no biological models
(ii) Has Instruments/equipment like microscope, Aquria and cages	Has no microscope, cages and Aquaria.

Any one related difference = 02 marks.

(a) (i) Solution A is Hypotonic Solution
B is Hypertonic Solution
C is Isotonic Solution.

(b) In Solution A cell bursted because cell Absorbed/gained water and cell has no cell wall

In Solution B cell shrank and shriveled because cell lost water

In Solution C cell did not change - because cell did not gain or lose water.

1 mark @ = 06 marks

(C) Characteristics of Respiratory Surface.

(i) It is moist to dissolve respiratory gases
(ii) It is thin to reduce diffusion distance.
(iii) They are well ventilated so that gases can pass through them easily.
(iv) They are Sorrounded by blood capellaries So that gases can be taken to and from the cells.

Any four points @1@mark = 04

Benefits of Insects

(i) They are used as food for Human being and other animals eg Grasshopper.

(ii) They are used for production of various substances eg Honeybee from Honey.

(iii) They are used in biological control of pest eg Lady beetles are used to kill Aphids.

(iv) They are used in agriculture as agent of pollination eg Butterfly.

(v) Some are used in production of medicine eg blister beetles that produce Cantharidin which used for treatment of urinogenital diseases

(v) Some improve soil aeration and soil fertility eg Termites break down Plant tissues

(vi) Some are used as tourist attraction eg Butterfly.

Any Six related points
Introduction = 1/2
Main body or @ = 12
Conclusion = 1/2
Total = 15 marks